## Homework

**Homework 6 — Study of a logarithmic function — 55 min.**

Consider the function $f$ defined on $]0;+\infty[$ by $f(x) = \frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of $f$ at the endpoints of its domain, study its sign, then draw up its table of variations.
2. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has exactly two solutions. Give for each a justified bracketing using calculated values.
4. Using the maximum of $f$, prove that $1 + \ln x \leq x$ for all $x > 0$. State the case of equality.